Midterm Exam 2

PhD Macroeconomics (Spring 2026)

Instructions. Show steps. You may explain in words where requested.

Question 1 (Asset pricing in an endowment economy)

[40 pts]

This question uses the notation from `AssetPricingEndowment.pdf`. In each period, the representative household receives fruit/dividend z. The household holds shares s of the fruit tree. The ex-dividend share price is p. Next period variables are denoted by primes: z', p', s', c'. The household can also trade a one-period treasury bill: hold a' bought at price q today and paying 1 next period.

Budget constraint (with both assets):

$$c + ps' + qa' \;\le\; (z+p)s + a. \tag{BC}$$

Assume equilibrium goods market clearing implies $c = z$ and equilibrium share supply implies $s = 1$.

(a) **Euler equations and returns.** [16 pts]

(i) Show the FOC for the treasury bill implies

$$u'(c)\, q \;=\; \beta\, \mathbb{E}_{z'|z}[u'(c')]\,. \tag{E-bond}$$

(ii) Show the FOC for shares implies

$$u'(c)\, p \;=\; \beta\, \mathbb{E}_{z'|z}[u'(c')\,(z'+p')]\,. \tag{E-share}$$

(iii) Define the **gross returns** exactly as in the slides:

$$e(z,z') \equiv \frac{z'+p'}{p}, \qquad R(z) \equiv \frac{1}{q}.$$

Rewrite (E-bond) and (E-share) as $1 = \mathbb{E}_{z'|z}[\text{SDF} \times \text{return}]$ using the slide's SDF.

(b) **Fundamental value (pricing by discounted dividends).** [14 pts]

Starting from (E-share) and using the transversality condition

$$\lim_{j\to\infty} \beta^j \mathbb{E}_t[u'(c_{t+j})\, p_{t+j}] = 0, \tag{TVC}$$

show that

$$u'(c_t)\, p_t = \mathbb{E}_t\left[\sum_{j=1}^{\infty} \beta^j u'(c_{t+j})\, z_{t+j}\right]. \tag{FV}$$

Then impose $c_t = z_t$ and rewrite p_t in terms of $\{z_{t+j}\}$ and marginal utilities (as in the slide).

(c) **Risk premium (covariance representation).** [10 pts]
Using the covariance identity from the slides, show that

$$\frac{\mathbb{E}_{z'|z}[e(z,z')] - R(z)}{R(z)} = -\operatorname{cov}_z\left(\underbrace{\beta\frac{u'(c')}{u'(c)}}_{\text{SDF}},\ e(z,z')\right). \tag{RP}$$

Give a one-sentence interpretation: when is the risk premium positive?

Question 2 (Asset pricing in a production economy) [30 pts]

This question uses the notation from `AssetPricingProduction.pdf`. A representative firm has labor-only technology

$$y = zn^{\alpha}, \qquad \alpha \in (0,1),$$

chooses labor n, pays wage w, and pays dividend $d = y - wn$. A representative household chooses (c, n, s') and has value function

$$V(s,z) = \max_{c\geq 0,\, s'\geq 0,\, n\geq 0} \Big\{ \log c + \psi(1-n) + \beta\mathbb{E}_{z'|z}[V(s',z')] \Big\},$$

subject to

$$c + ps' \leq (d+p)s + wn.$$

In equilibrium: goods market clears $c = y$ and share market clears $s = 1$.

(a) **Firm: wages and dividends.** [10 pts]
Show the firm FOC implies $w = \alpha z n^{\alpha-1}$. Show the dividend is $d = (1-\alpha)y$.

(b) **Household FOC for labor.** [8 pts]
Show the household FOC implies

$$\frac{w}{c} = \psi \quad \Rightarrow \quad w = \psi c. \tag{L}$$

Use equilibrium $c = y$ and the firm wage equation to solve for

$$n = \frac{\alpha}{\psi} \qquad \text{and} \qquad y = z\left(\frac{\alpha}{\psi}\right)^{\alpha}.$$

(c) **Share Euler equation and constant p/y.** [12 pts]
From the household FOC for s', show:

$$\frac{1}{c}\,p = \beta\,\mathbb{E}_{z'|z}\left[\frac{1}{c'}(d'+p')\right]. \tag{S}$$

Using $c = y$ and $d' = (1-\alpha)y'$, show:

$$\frac{p}{y} = \beta(1-\alpha) + \beta\,\mathbb{E}_{z'|z}\left[\frac{p'}{y'}\right].$$

Now guess $\frac{p}{y} \equiv \Lambda$ is constant and solve for Λ. State the implied comovement between p and GDP y.

Question 3 (RBC: facts/HP filter, BGP, deflation, measurement) [30 pts]

This question uses the notation from the RBC Topic 1A and Topic 1BC slides.

(a) **HP filter (objective and notation).** [12 pts]
Given data $\{y_t\}_{t=1}^T$, the HP filter chooses a trend $\{\tau_t\}_{t=1}^T$. Write the HP minimization problem exactly as in the slides:

$$\min_{\{\tau_t\}_{t=1}^T} \frac{1}{T}\sum_{t=1}^{T}(y_t-\tau_t)^2+\frac{\lambda}{T-1}\sum_{t=2}^{T-1}\Big[(\tau_{t+1}-\tau_t)-(\tau_t-\tau_{t-1})\Big]^2,$$

and define the cycle $\tilde{y}_t$. In one sentence, what does λ control?

(b) **Balanced growth logic.** [8 pts]
In the one-sector growth model with labor-augmenting progress,

$$Y_t = zF(K_t, n_t X_t), \qquad \frac{X_{t+1}}{X_t} = \gamma_x \geq 1,$$

and define gross growth rates $\gamma_v(t) \equiv v_{t+1}/v_t$. Assume z and n_t are constant on the BGP. Explain briefly why a BGP requires

$$\gamma_y = \gamma_c = \gamma_i = \gamma_k \qquad \text{and} \qquad \gamma_y = \gamma_k = \gamma_x.$$

(c) **Growth-deflated constraints.** [6 pts]
Define detrended (growth-deflated) variables

$$y_t \equiv \frac{Y_t}{X_t}, \quad c_t \equiv \frac{C_t}{X_t}, \quad i_t \equiv \frac{I_t}{X_t}, \quad k_t \equiv \frac{K_t}{X_t}.$$

Write the growth-deflated resource constraint and capital accumulation constraint as in the slides:

$$y_t \geq c_t + i_t, \qquad \gamma_x k_{t+1} \leq (1-\delta)k_t + i_t.$$

(d) **Solow residual / TFP measurement.** [4 pts]
With Cobb–Douglas production,

$$Y_t = A_t K_t^{\alpha}(n_t X_t)^{1-\alpha},$$

show that

$$\log A_t = \log Y_t - \alpha \log K_t - (1-\alpha)\log n_t - (1-\alpha)\log X_t.$$

Then compute A_t for: $Y_t = 100$, $K_t = 300$, $n_t = 0.30$, $X_t = 2$, $\alpha = 0.34$.